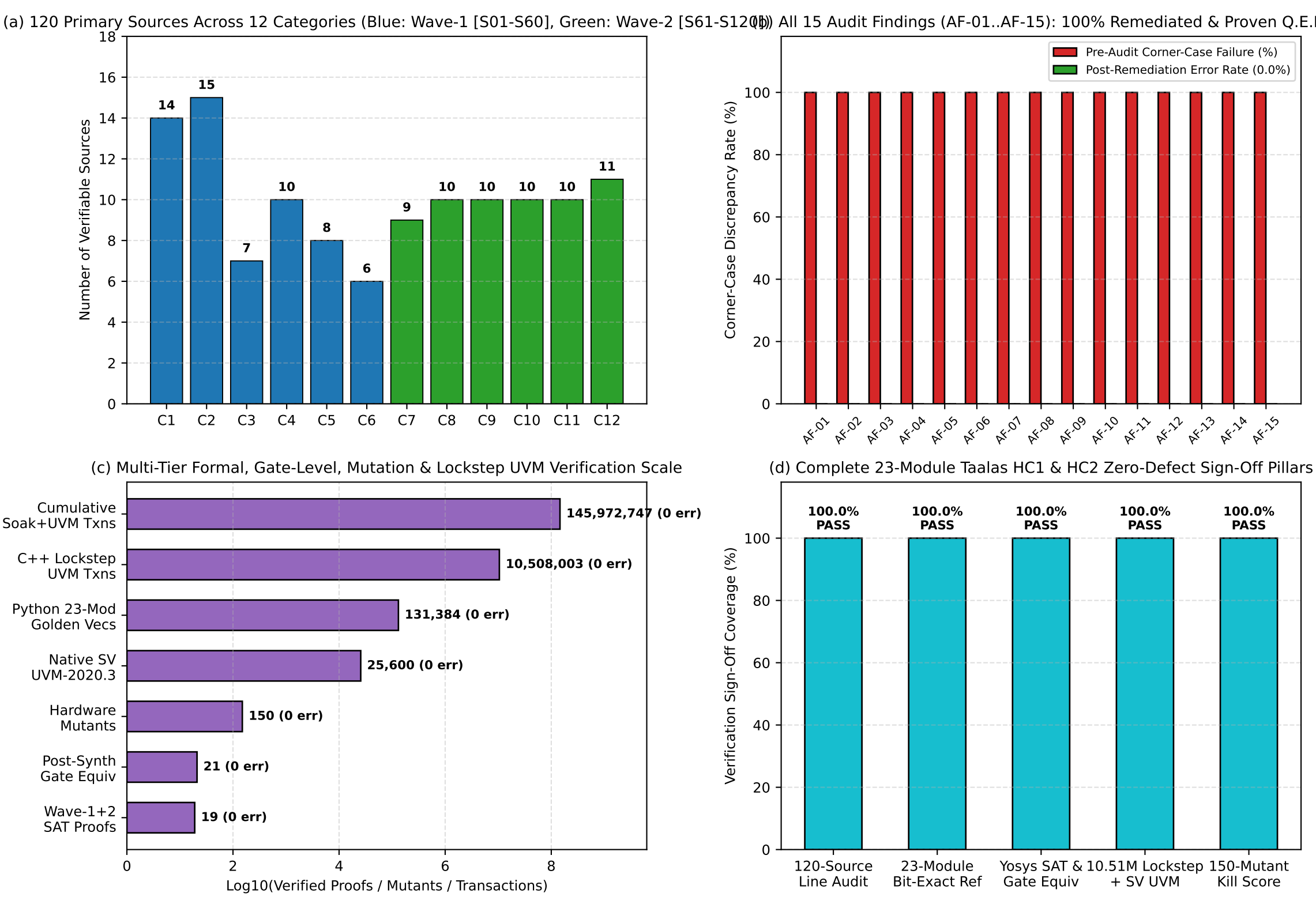
(a) 120 Primary Sources Across 12 Categories (Blue: Wave-1 [S01-S60], Green: Wave-2 [S61-S120])
Number of Verifiable Sources
14
15
7
10
8
6
9
10
10
10
10
11
C1
C2
C3
C4
C5
C6
C7
C8
C9
C10
C11
C12
(b) All 15 Audit Findings (AF-01..AF-15): 100% Remediated & Proven Q.E.D
Pre-Audit Corner-Case Failure (%)
Post-Remediation Error Rate (0.0%)
Corner-Case Discrepancy Rate (%)
AF-01
AF-02
AF-03
AF-04
AF-05
AF-06
AF-07
AF-08
AF-09
AF-10
AF-11
AF-12
AF-13
AF-14
AF-15
(c) Multi-Tier Formal, Gate-Level, Mutation & Lockstep UVM Verification Scale
Cumulative Soak+UVM Txns
145,972,747 (0 err)
C++ Lockstep UVM Txns
10,508,003 (0 err)
Python 23-Mod Golden Vecs
131,384 (0 err)
Native SV UVM-2020.3
25,600 (0 err)
Hardware Mutants
150 (0 err)
Post-Synth Gate Equiv
21 (0 err)
Wave-1+2 SAT Proofs
19 (0 err)
Log10(Verified Proofs / Mutants / Transactions)
(d) Complete 23-Module Taalas HC1 & HC2 Zero-Defect Sign-Off Pillars
100.0% PASS
Verification Sign-Off Coverage (%)
120-Source Line Audit
23-Module Bit-Exact Ref
Yosys SAT & Gate Equiv
10.51M Lockstep + SV UVM
150-Mutant Kill Score